PHD22 Macro labour: Problem set

Rob Włodarski

April 2026

Question 1: The DMP model

I follow the exposition from Hagedorn and Manovskii (2008) wherever possible. The Bellman equations that govern the model are:

$$J(p) = p - w(p) + \beta(1-s)\mathbb{E}_p[J(p')], \tag{1a}$$

$$V(p) = -c + \beta q[\theta(p)]\mathbb{E}_p[J(p')], \tag{1b}$$

$$U(p) = b + \beta\langle f[\theta(p)]\mathbb{E}_p[W(p')] + \{1 - f[\theta(p)]\}\mathbb{E}_p[U(p')]\rangle, \tag{1c}$$

$$W(p) = w(p) + \beta\{(1-s)\mathbb{E}_p[W(p')] + s\,\mathbb{E}_p[U(p')]\}. \tag{1d}$$

I use the matching function from Den Haan et al. (2000) :

$$m(u,v) = \frac{uv}{(u^l + v^l)^{\frac{1}{l}}}, \tag{2}$$

which gives the following contact rates:

$$q[\theta(p)] = \left[1 + \theta(p)^l\right]^{-\frac{1}{l}}, \tag{3a}$$

$$f[\theta(p)] = \left[1 + \theta(p)^{-l}\right]^{-\frac{1}{l}}. \tag{3b}$$

Free entry condition

The **free entry condition** stipulates $V(p) = 0$ which translates to:

$$c = \beta q[\theta(p)]\mathbb{E}_p[J(p')], \tag{4a}$$

$$J(p) = p - w(p) + \frac{(1-s)c}{q[\theta(p)]}. \tag{4b}$$

The free entry condition can be used to derive the job posting condition. In the steady state, where we have only one price, $p = p^\star$, the conditions collapse to:

$$c = \beta q(\theta^\star) J^\star \implies J^\star = \frac{c}{\beta q(\theta^\star)} \tag{5a}$$

$$J^\star = p^\star - w^\star + \beta(1-s)J^\star \implies J^\star = \frac{p^\star - w^\star}{1 - \beta(1-s)}. \tag{5b}$$

Steady state job creation condition

Setting $r \equiv \frac{1}{\beta} - 1$, the **job creation condition** becomes:

$$w^\star = p^\star - \frac{r+s}{q(\theta^\star)} c. \tag{6}$$

Nash bargaining and surplus sharing

A worker and a firm split the surplus, with workers taking γ:

$$W(p) = \gamma \{J(p) + W(p) - U(p)\}, \tag{7a}$$

$$J(p) = (1-\gamma)\{J(p) + W(p) - U(p)\}. \tag{7b}$$

Using the surplus sharing condition, I solve for the closed-form solution of $w(p)$:

$$[W]: \quad W(p) - \beta \mathbb{E}_p[W(p')] = w(p) - \beta s \mathbb{E}_p[W(p') - U(p')], \tag{8a}$$

$$[U]: \quad U(p) - \beta \mathbb{E}_p[U(p')] = b + \beta f[\theta(p)] \mathbb{E}_p[W(p') - U(p')], \tag{8b}$$

which can be combined:

$$W(p) - U(p) = w(p) - b + \beta \{1 - s - f[\theta(p)]\} \mathbb{E}_p[W(p') - U(p')]. \tag{8c}$$

Now, by Nash bargaining and free entry, we must have that:

$$W(p) - U(p) = \frac{\gamma}{1-\gamma} J(p) = \frac{\gamma}{1-\gamma}\left[p - w(p) + \frac{(1-s)c}{q[\theta(p)]}\right], \tag{8d}$$

$$\mathbb{E}_p[W(p') - U(p')] = \frac{\gamma}{1-\gamma} \mathbb{E}_p[J(p)] = \frac{\gamma}{1-\gamma} \frac{c}{\beta q[\theta(p)]}. \tag{8e}$$

The last three conditions can be combined and rearranged to solve for the wage.

Equilibrium wage

By combining the free entry condition with Nash bargaining, I obtain the closed form wage:

$$w(p) = \gamma[p + \theta(p) c] + (1-\gamma) b. \tag{9}$$

Beveridge curve

From the law of motion of employment:

$$n' = (1-s) n + m(v, u) \tag{10a}$$

I obtain the steady state Beveridge curve:

$$u^\star = \frac{s}{f+s}. \tag{10b}$$

Question 2: The steady state algorithm

Algorithm 1 sets out the standard procedure of calibrating a parameter, in this case c, in the DMP model.

Algorithm 1: Calibration of vacancy posting cost to target unemployment

Input: Parameters $(l, \gamma, b, \beta, s, p^\star)$, target unemployment rate $u^\star$
Output: Calibrated $c^\star$, equilibrium objects $(\theta^\star, w^\star, q^\star, f^\star, u^\star)$

```
repeat
    // Inner steady-state problem given c
    Use the bisection (or alternative) algorithm;
    repeat
        Compute implied tightness θ from Equation (3a) ;
        Compute implied wage w from Equation (9) ;
        Compute θ̃ from the job-creation condition, Equation (6) ;
        Update q by bisection on the residual |q^{n+1} − q^n|;
    until |q^{n+1} − q^n| < ε_q;
    // Evaluate calibration target
    Compute job-finding rate f from the matching function in Equation (3b);
    Compute steady-state unemployment u from Equation (10b);
    Update c by bisection on the residual u − u*;
until |u − u*| < ε_κ;
```

However, I compute c directly, as the model allows to construct c as a function of u and remaining parameters. Start by combining the job creation condition and the equilibrium wage from Equations (6) and (9):

$$\gamma\left(p^\star + \theta^\star c\right) + \left(1-\gamma\right) b = p^\star - \frac{r+s}{q(\theta^\star)} c \tag{11a}$$

and rearrange a little bit:

$$c = \frac{\left(1-\gamma\right)}{\frac{r+s}{q(\theta^\star)} + \gamma\theta^\star}. \tag{11b}$$

$\theta^\star$ and $q(\theta^\star)$ can be obtained from the Beveridge curve in Equation (10b):

$$u = \frac{s}{f(\theta^\star) + s}. \tag{11c}$$

"Algorithm" 2 fleshes out the fast way to do it (I abstract from computing a large closed form solution).

Algorithm 2: Calibration of vacancy posting cost (closed form)

Input: Parameters $(l, \gamma, b, \beta, s, p^\star)$, target unemployment rate $u^\star$
Output: Calibrated $c^\star$, equilibrium objects $(\theta^\star, w^\star, q^\star, f^\star)$

```
// Pin down the flow values and tightness
Compute job-finding rate f* from the Beveridge curve, Equation (10b);
Compute tightness θ* by inverting Equation (3b);
Compute job-filling rate q* from Equation (3a);
// Free entry and Nash bargaining pin down c
Jointly solve Equation (6) and Equation (9) for c*;
Compute equilibrium wage w* from Equation (9);
```

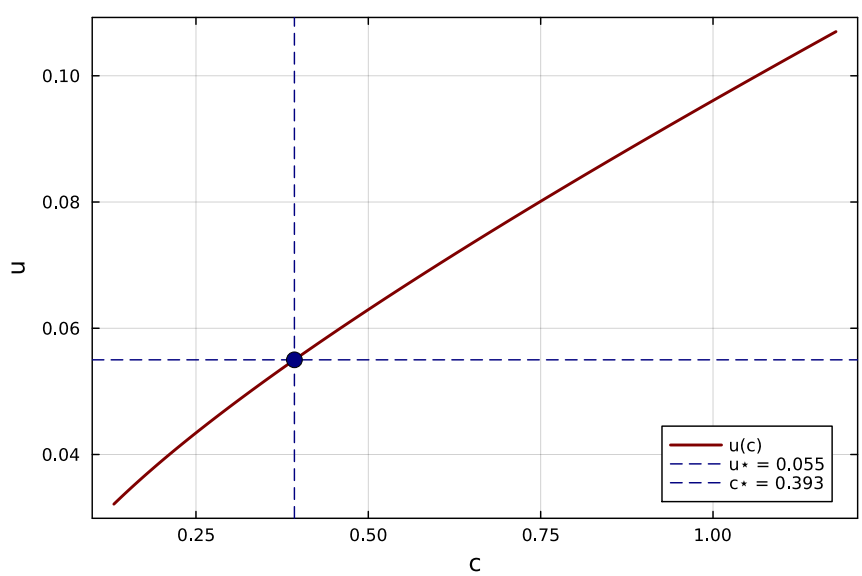
Question 3: Calibrating c

Table 1 summarises the calibration exercise. The calibrated vacancy cost is relatively high, at around 39% of the productivity value. Figure 1 indicates that the parameter is well-identified.

Table 1: Steady-state calibration

Description	**Symbol**	**Value**
Parameters		
Home production	b	0.4000
Discount factor	β	0.9992
Separation rate	s	0.0081
Bargaining power	γ	0.7000
Matching elasticity	l	0.4070
Steady-state productivity	$p^\star$	1.0000
Calibration		
Target unemployment rate	$u^\star$	0.0550
Calibrated vacancy cost	$c^\star$	0.3928
Equilibrium objects		
Labour market tightness	$\theta^\star$	0.5997
Wage	$w^\star$	0.9849
Job-filling rate	$q^\star$	0.2321
Job-finding rate	$f^\star$	0.1392
Unemployment rate	$u^\star$	0.0550
Firm value	$J^\star$	1.6938

Figure 1: Matching c to u



Question 4: Solving the model with aggregate uncertainty

I use the Rouwenhorst method to discretise the productivity process. This is more fitting for the highly persistent process, with $\rho \simeq 0.99$, than the Tauchen method. I set 15 as the number of gridpoints. Table 2 summarises the values of all equilibrium objects at the minimum and maximum productivity grids. Predictably, the job-finding rate varies quite little with the productivity process.

Table 2: Equilibrium objects across productivity states

Description	**Symbol**	**SS**	**Min** p	**Max** p
Productivity	p	1.0000	0.9874	1.0128
Labour market tightness	θ	0.5997	0.5870	0.6125
Wage	w	0.9849	0.9725	0.9974
Job-filling rate	q	0.2321	0.2343	0.2299
Job-finding rate	f	0.1392	0.1375	0.1408
Firm value	J	1.6938	1.6776	1.7101

Question 5: Towards Hagedorn and Manovskii (2008)

Hagedorn and Manovskii (2008)'s Table 4 presents simulation statistics based on the simulation of monthy data, detreneded using the HP filter with a smoothing parameter of $1,600$.[1] My replication is summarised in Table 3 My results are based on a 100 year simulation ($T = 5200$ periods, with a 20-year burnin).

Table 3: Results from the calibrated model

	u	v	v/u	p
Standard deviation	0.003	0.003	0.006	0.004
Autocorrelation	1.000	0.999	1.000	1.000
Correlation matrix				
u	1.000	-0.946	-0.987	-0.987
v	—	1.000	0.986	0.986
v/u	—	—	1.000	1.000
p	—	—	—	1.000

Naturally, I expect the values to depend on three types of variables:

- ρ and σ: If one adjusts the productivity process, the business cycle properties will change, too.
- T: Setting a too small value of T may mean that we don't visit far ends of each tail.
- N_p: A coarse grid may mean that we fail to capture the correlation structure of the key labour variables. Too many grid points means that the tails are too rarely visited (unless T is adjusted).

For example, I check what happens when the number of grid points is adjusted. Figures 2 and 3 illustrate how the key correlations and standard deviations change for different numbers of grid points. Unsurprisingly, we cannot solve the Shimer puzzle this way - the volatility of unemployment is quite far from 0.125.

[1] Am I understading it right? I thought a value of $1,600$ was reserved for the quarterly business cycle variables.

Figure 2: Correlation sensitivity to changes in N_p

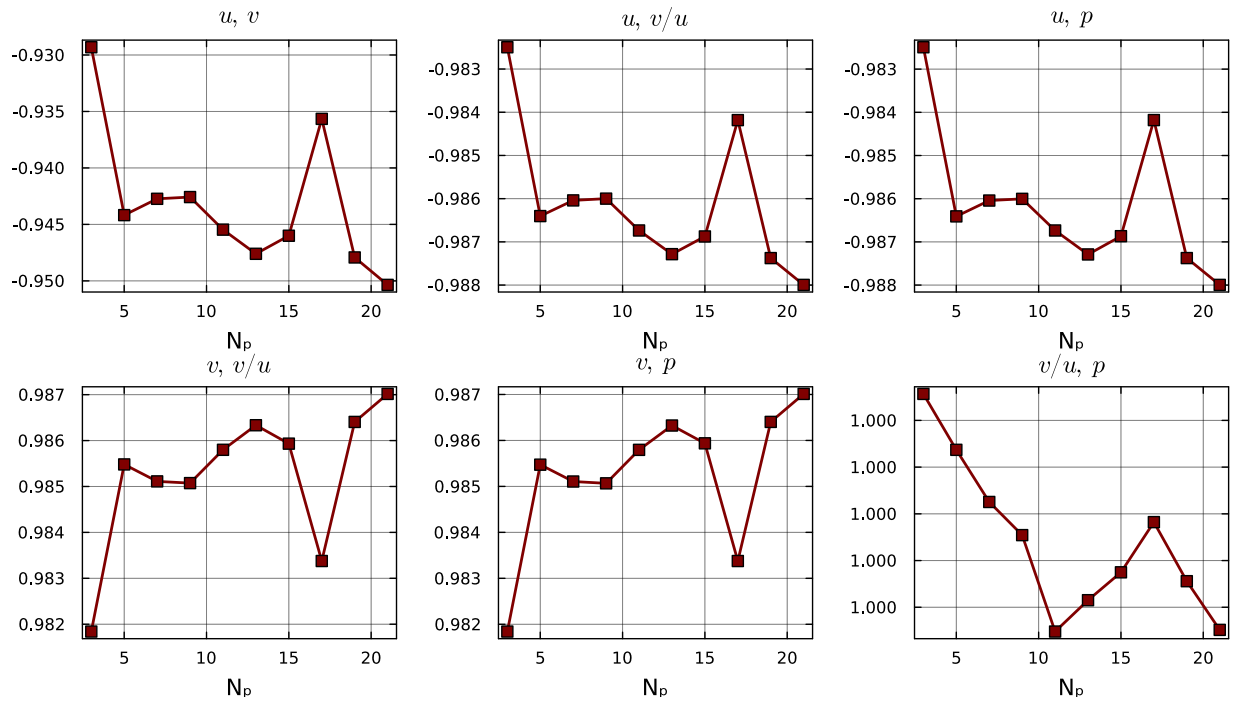
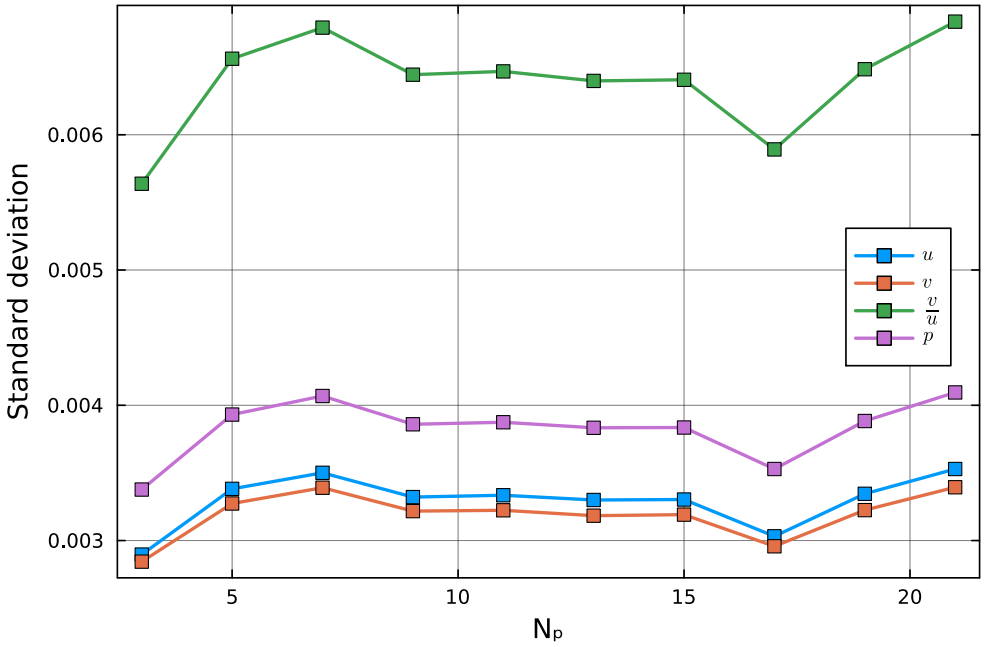


Figure 3: Standard deviation sensitivity to changes in N_p



Question 6: Hagedorn and Manovskii (2008)'s parameters

Table 4: Steady-state calibration with alternative (b, γ)

Description	Symbol	Value
Parameters		
Home production	b	0.9400
Discount factor	β	0.9992
Separation rate	s	0.0081
Bargaining power	γ	0.0520
Matching elasticity	l	0.4070
Steady-state productivity	$p^\star$	1.0000
Calibration		
Target unemployment rate	$u^\star$	0.0550
Calibrated vacancy cost	$c^\star$	0.8161
Equilibrium objects		
Labour market tightness	$\theta^\star$	0.5997
Wage	$w^\star$	0.9686
Job-filling rate	$q^\star$	0.2321
Job-finding rate	$f^\star$	0.1392
Unemployment rate	$u^\star$	0.0550
Firm value	$J^\star$	3.5195

Table 5: Equilibrium objects across productivity states

Description	Symbol	SS	Min p	Max p
Productivity	p	1.0000	0.9874	1.0128
Labour market tightness	θ	0.5997	0.4748	0.7315
Wage	w	0.9686	0.9626	0.9748
Job-filling rate	q	0.2321	0.2570	0.2119
Job-finding rate	f	0.1392	0.1220	0.1550
Firm value	J	3.5195	3.1750	3.8587

Table 6: Results from the calibrated model (Hagedorn and Manovskii, 2008)

	u	v	v/u	p
Standard deviation	0.034	0.033	0.066	0.004
Autocorrelation	1.000	0.999	1.000	1.000
Correlation matrix				
u	1.000	-0.945	-0.987	-0.986
v	—	1.000	0.986	0.985
v/u	—	—	1.000	0.999
p	—	—	—	1.000

Question 7: Experimentation with (b, γ)

Figure 4: Experiment: Steady state equilibrium values

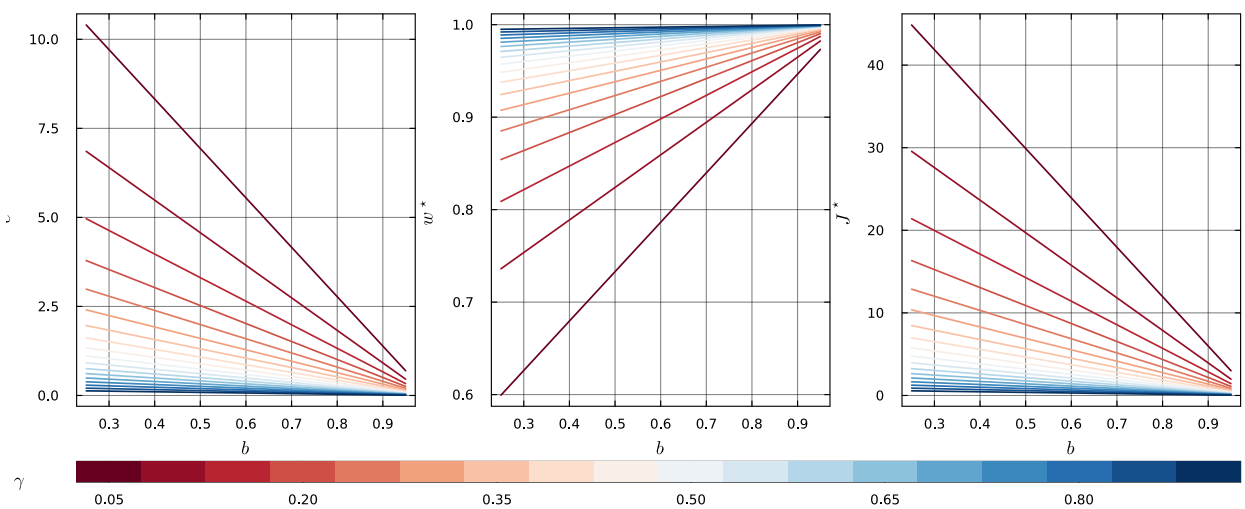


Figure 5: Experiment: Aggregate uncertainty ranges

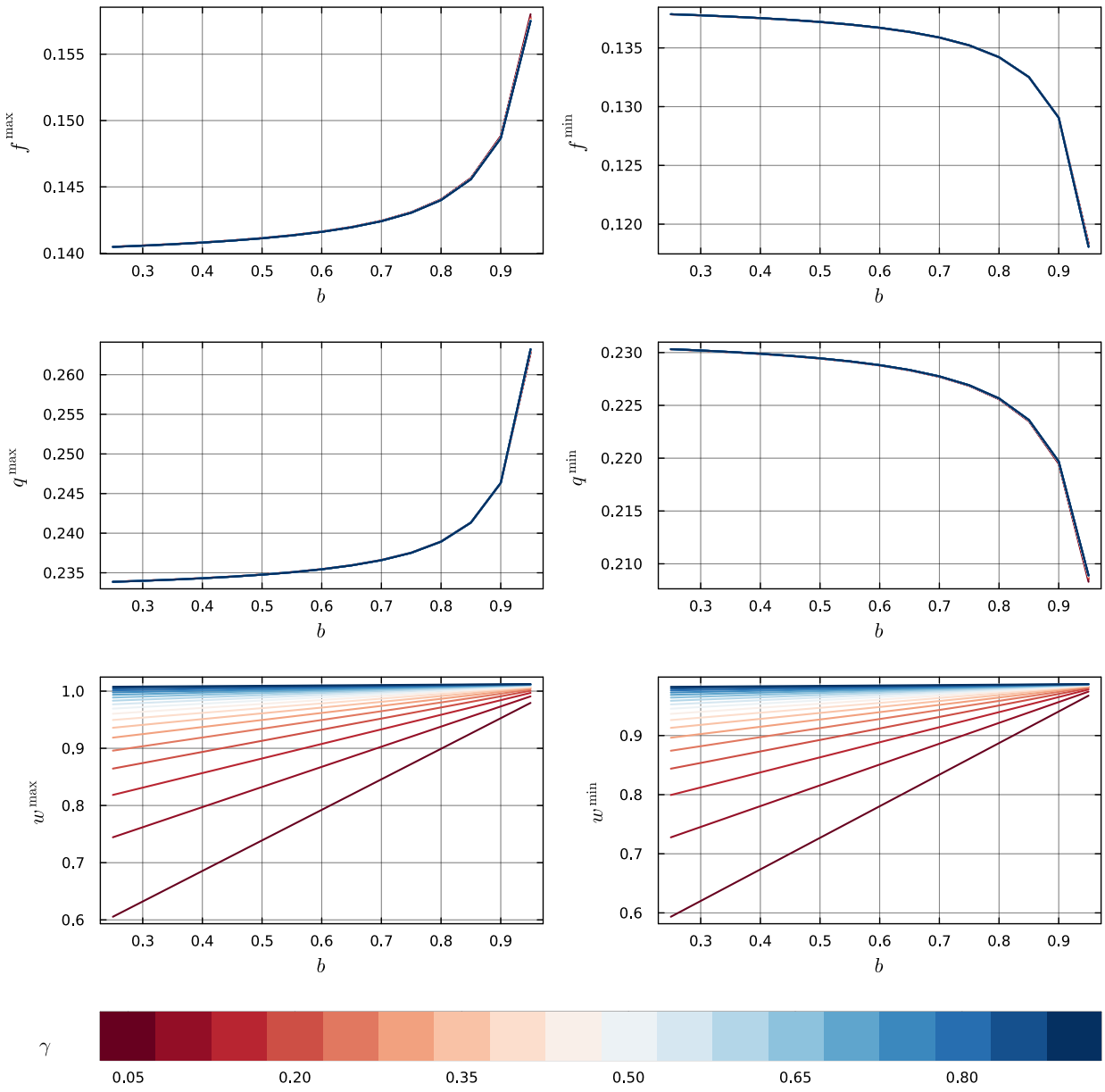
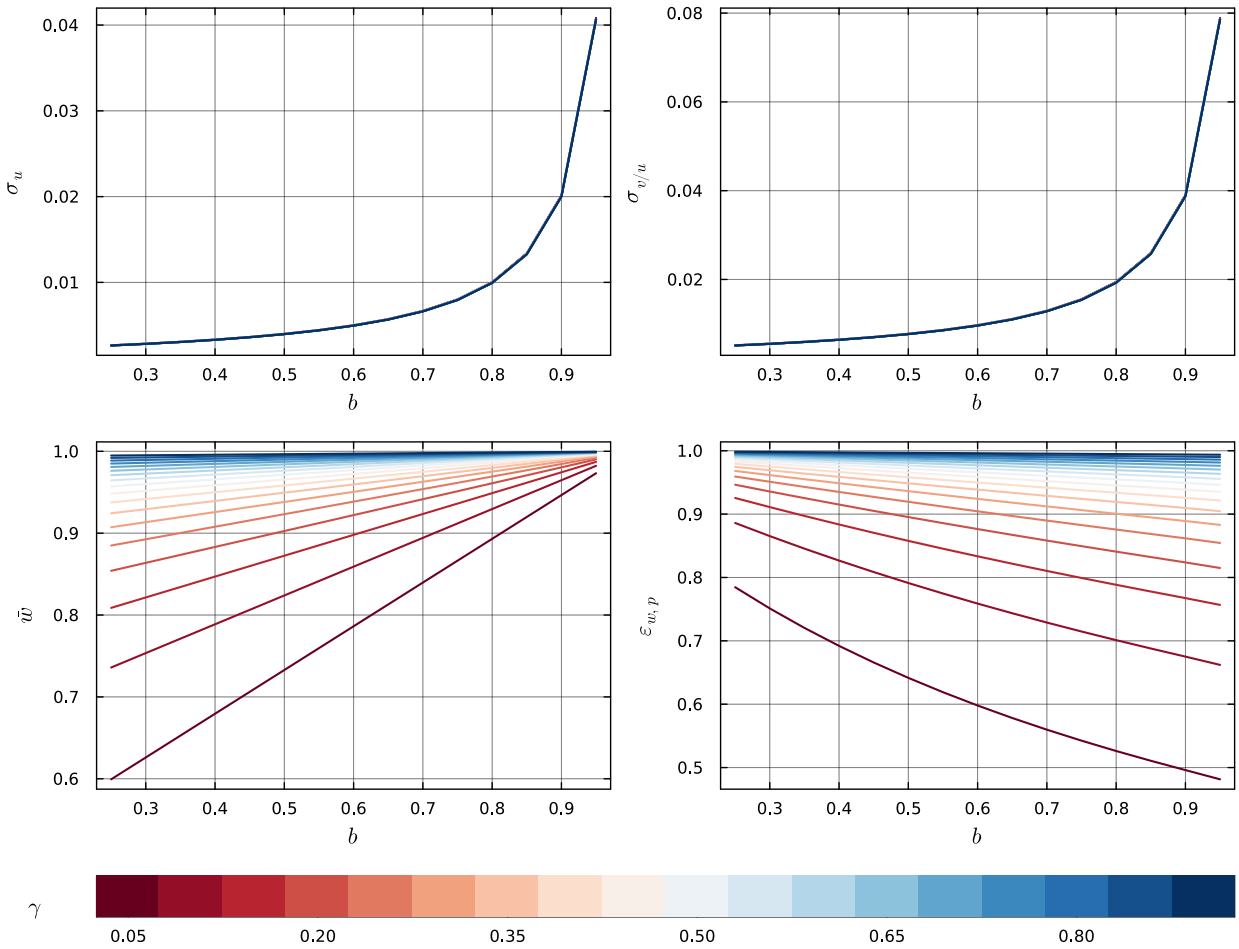


Figure 6: Experiment: Simulation results



References

Den Haan, Wouter J, Garey Ramey, and Joel Watson (June 2000). "Job Destruction and Propagation of Shocks". en. In: *American Economic Review* 90.3, pp. 482–498. ISSN: 0002-8282. DOI: 10.1257/aer.90.3.482. URL: https://pubs.aeaweb.org/doi/10.1257/aer.90.3.482 (visited on 04/21/2026).

Hagedorn, Marcus and Iourii Manovskii (Aug. 2008). "The Cyclical Behavior of Equilibrium Unemployment and Vacancies Revisited". en. In: *American Economic Review* 98.4, pp. 1692–1706. ISSN: 0002-8282. DOI: 10.1257/aer.98.4.1692. URL: https://pubs.aeaweb.org/doi/10.1257/aer.98.4.1692 (visited on 04/21/2026).